IBM Z DATATHON 2026 · THEME 1

SENTINEX

Industrial AI Safety Intelligence and Risk Governance

Anticipate the risk. Act before it escalates.

NEXGAURD AI · Built on IBM LinuxONE (s390x)

The problem

>72%

False alarm rate

Conventional wearable safety devices overwhelm responders with alert fatigue.

Cloud

Unpredictable latency

Cloud-reliant pipelines add non-deterministic delay and move sensitive health data off-site.

Black box

No accountability

The EU AI Act and GDPR Article 9 do not allow unconstrained ML authority over emergency escalation.

SENTINEX answers with edge-to-mainframe inference on IBM LinuxONE, deterministic guardrails and an auditable decision trail.

How SENTINEX works

01

Edge ingress

100 Hz accelerometer, ECG/PPG and acoustics in a W=100 window

02

Dual-engine ML

Conformal Isolation Forest plus calibrated HistGradientBoosting

03

Guardrails

Hard limits: HR above 190 or below 35 bpm, impact above 6.5 g

04

Consent gate

GDPR Art. 9 state machine with the Art. 9(2)(c) vital-interest exception

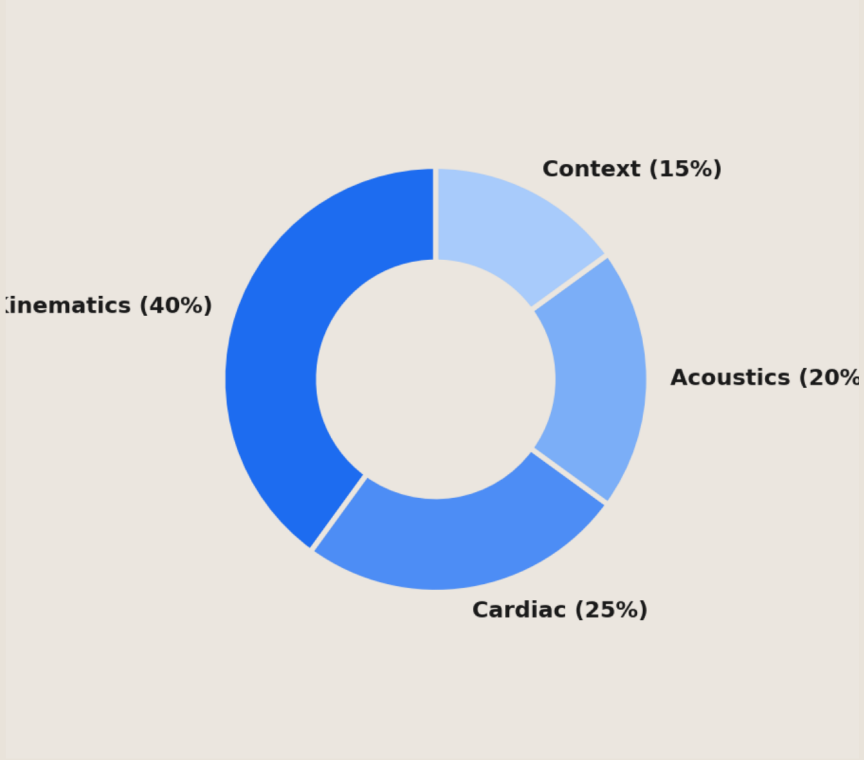
05

Audit ledger

SHA-256 hash chain of every decision for EU AI Act audits

$$\text{Conformal cutoff } \hat{q} = -0.0212 \cdot P(\text{miss}) \leq 0.05 \cdot 95\% \text{ coverage guarantee}$$

Explainable risk scoring



Every alert shows why it fired

Z-score attribution splits each risk score across kinematics (40%), cardiac (25%), acoustics (20%) and baseline context (15%).

Drift is monitored live

Population Stability Index: stable below 0.10, recalibration warning from 0.10 to 0.25, conservative fallback above 0.25.

Benchmark results

MODEL	F1	AUROC	LATENCY (s390x)
Conformal Isolation Forest (ours)	**0.941**	**0.978**	**5.41 ms**
HistGradientBoosting	0.923	0.962	6.20 ms
MLP Autoencoder	0.845	0.891	12.30 ms
One-Class SVM (RBF)	0.812	0.854	18.50 ms
Static threshold baseline	0.580	0.612	0.80 ms

Tested on 10,000 synthetic and physiological stress sequences. The static threshold is fastest but misses the most events (F1 0.580).

Training on LinuxONE



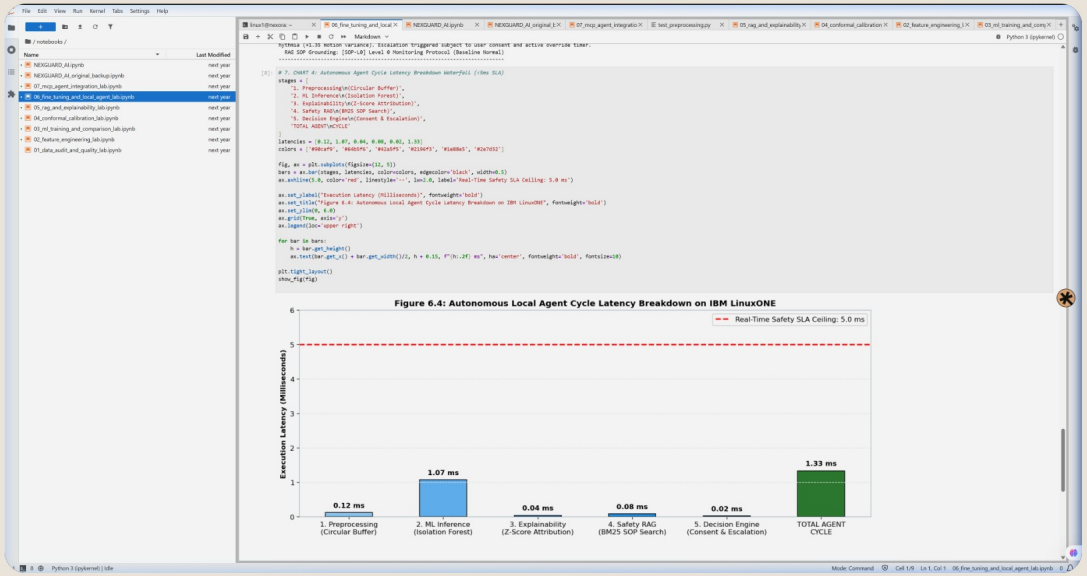
99.67%

Validation accuracy of the HistGradientBoosting risk adapter on a 30% stratified holdout.

Training error keeps falling as boosting iterations grow.

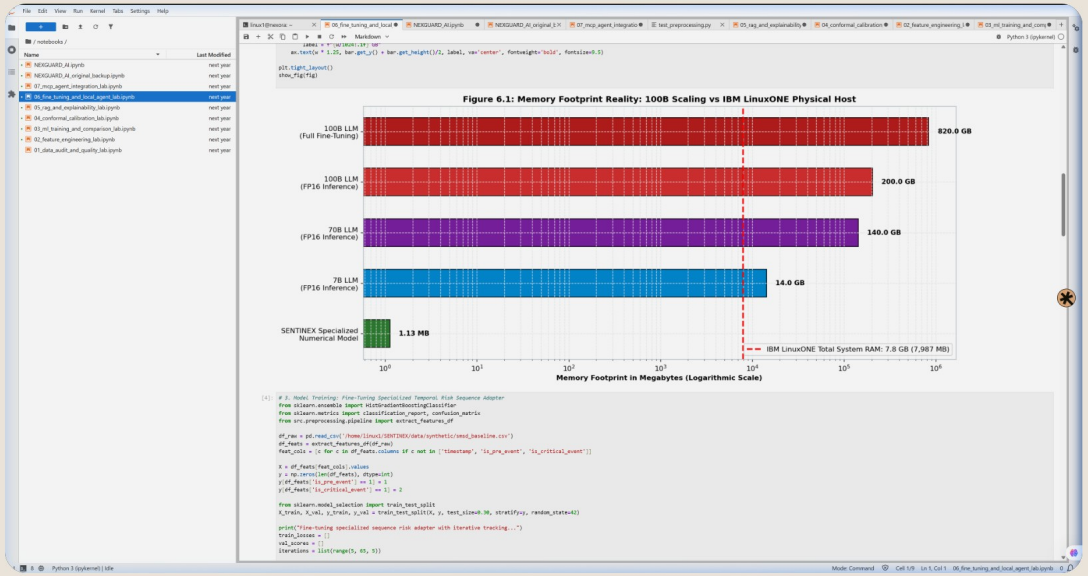
Figure 6.2 · notebook 06, 5 to 60 boosting iterations

Speed and footprint



1.33 ms agent cycle

Measured in the notebook against a 5 ms real-time safety SLA.

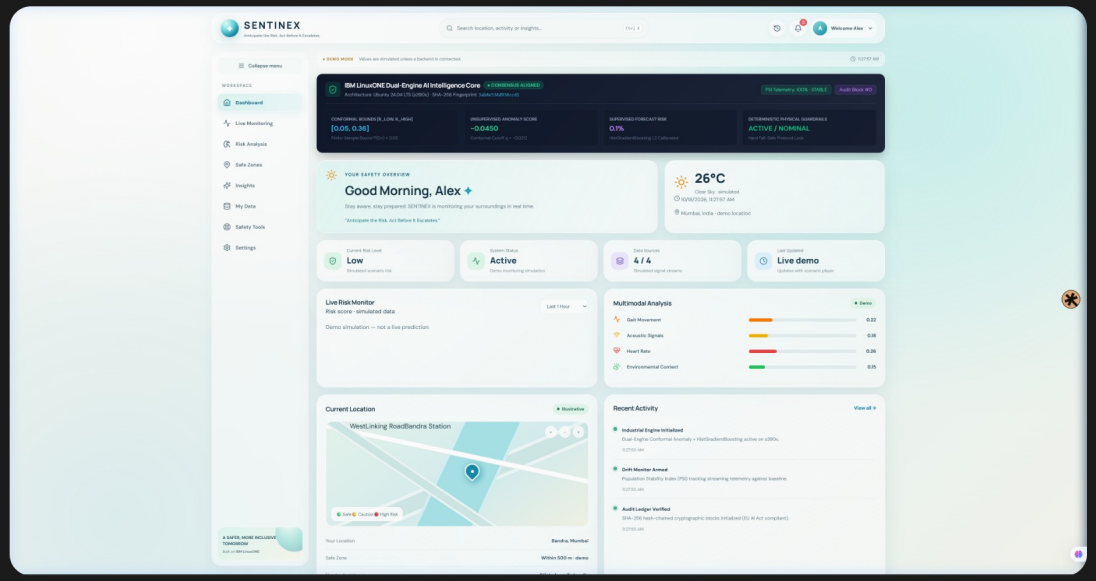


1.13 MB model

A 7B LLM needs 14 GB; the LinuxONE host has 7.8 GB of RAM.

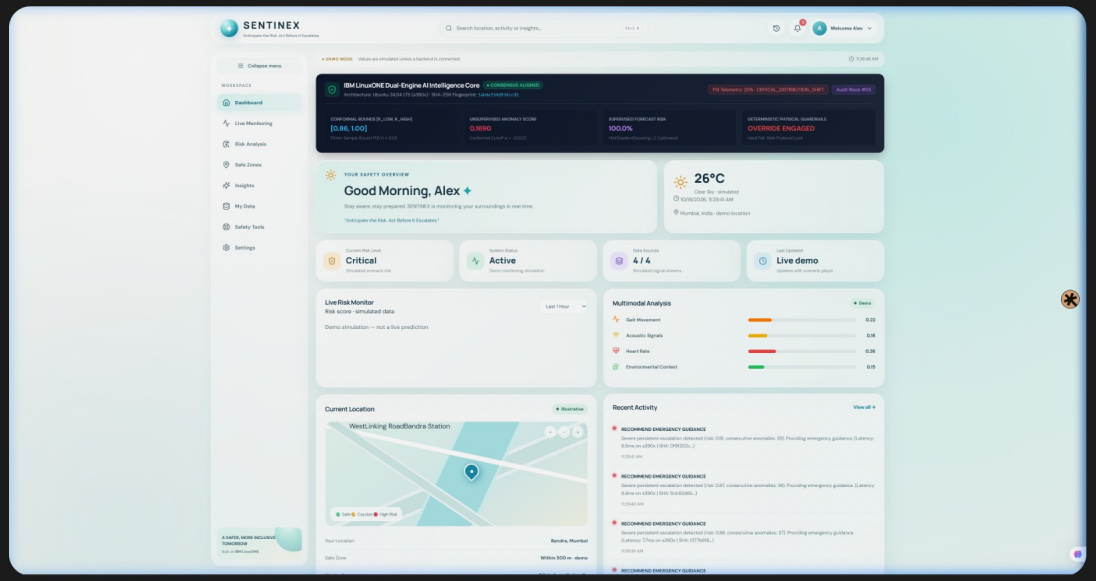
Figures 6.4 and 6.1 · IBM LinuxONE (s390x)

Live product demo



Nominal

Conformal bounds [0.05, 0.36], PSI stable, guardrails nominal.

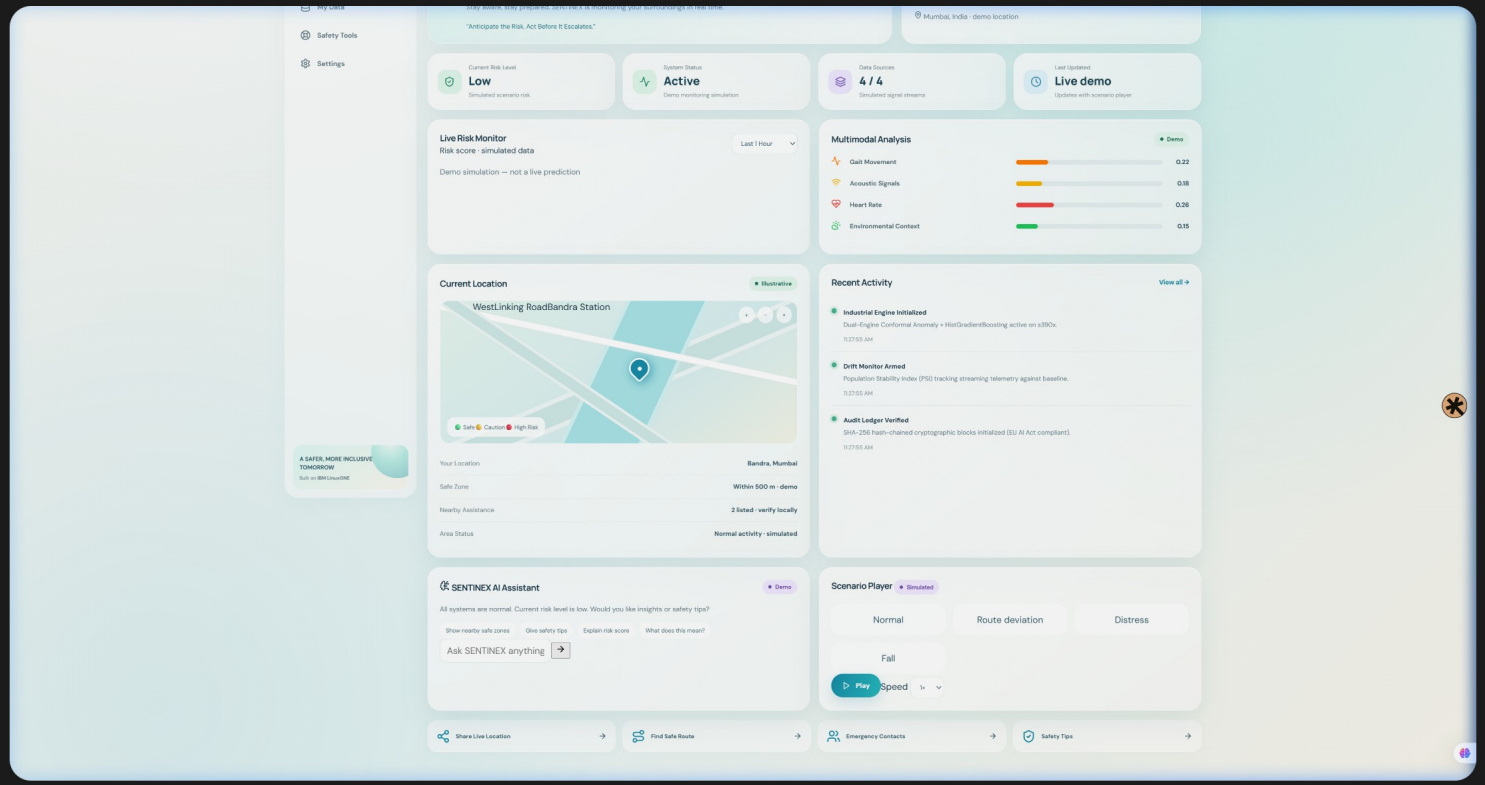


Injected distress

Bounds [0.86, 1.00], guardrail override engaged, audit block #55 written.

Demo mode: scenario player with simulated telemetry · React + Vite frontend, FastAPI backend

Dashboard modules



Multimodal analysis

Gait, acoustics, heart rate and environment scored side by side.

Scenario player

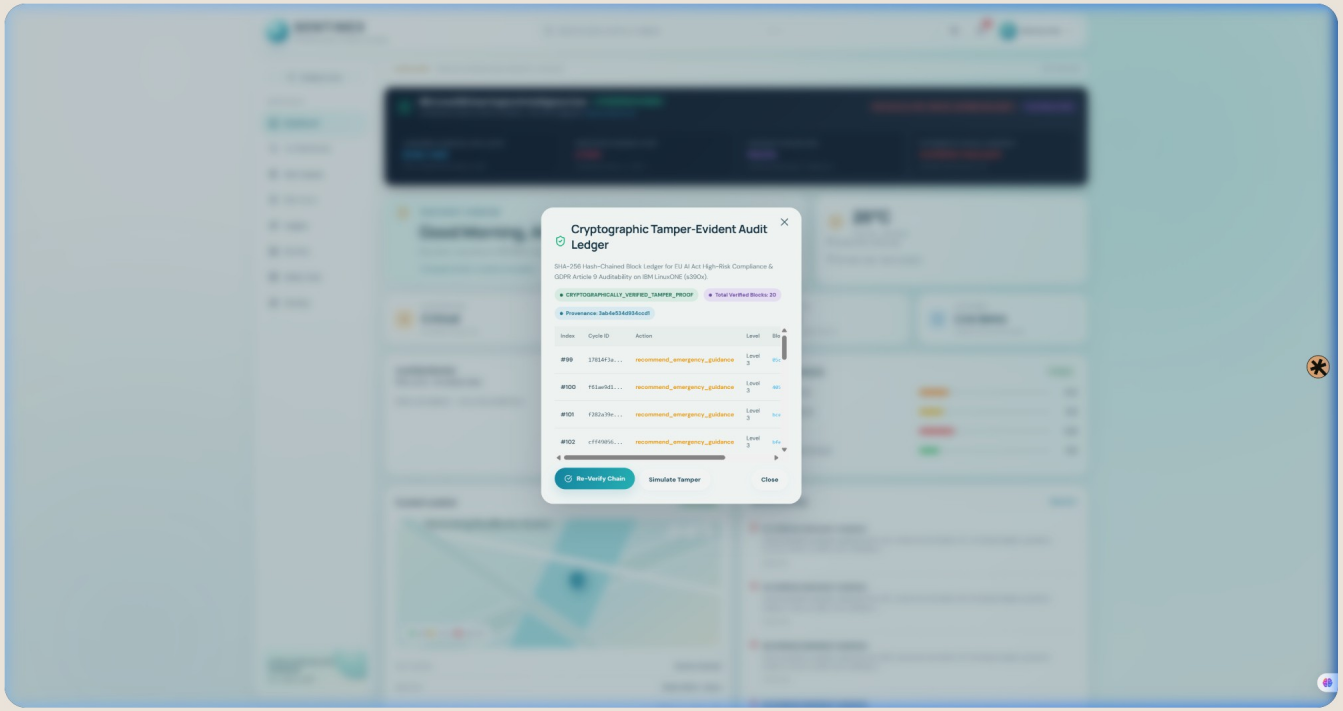
Normal, route deviation, distress and fall scenarios on demand.

AI assistant

Explains the risk score and suggests safety actions.

Demo mode: simulated telemetry

Tamper-evident audit ledger



Each block records

- Index and timestamp
- Previous block hash
- Decision and attributions
- Consent state
- Current SHA-256 hash

Re-verify the chain or simulate a tamper live.

EU AI Act high-risk audit trail · provenance 3ab4e534d934ccd1

Trust and compliance

≤ 5%

Conformal miss bound

Distribution-free coverage guarantee on anomaly alerts.

53 / 53

Unit tests passing

Calibration, rolling stats, overrides and ledger verification.

PSI > 0.25

Drift fallback

Critical distribution shift triggers a conservative fallback.

1.13 MB

Model footprint

Runs on a 7.8 GB LinuxONE host with no GPU dependency.

WHY IBM LINUXONE

Auditable, on-premises safety AI in under 20 ms

Under 20 ms full-pipeline target, 1.33 ms measured agent cycle

Zero GPU dependency, linear scaling across s390x threads

Every decision hash-chained for EU AI Act review

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